

Transaction Processing, ACID Properties & Concurrency Control

Subject: CS402 | Unit 4: Transaction Management

1. Executive Overview & Syllabus Scope

In-depth notes on ACID properties, Conflict Serializability, Precedence Graphs, Two-Phase Locking 2PL, Deadlock Detection, a

2. Key Architectural Invariants & Core Topics

â€ Lecture Modules & Key Topics

- Unit 4: Transaction Processing & Concurrency Protocols
- 1. ACID Properties: Atomicity, Consistency, Isolation, Durability breakdown
- 2. Serializability: Conflict Serializability, View Serializability, and Precedence Graph testing
- 3. Lock-Based Protocols: Shared S and Exclusive X locks; Two-Phase Locking 2PL
- 4. Strict 2PL vs Rigorous 2PL: Cascading abort prevention and serializable schedules
- 5. Deadlock Handling: Wait-Die vs Wound-Wait timestamp schemes and Wait-For Graphs
- 6. Log-Based Recovery: Write-Ahead Logging WAL and ARIES recovery algorithm

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.